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AI Carbon Credit Verifier

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***Abstract*- Carbon credit markets are at the heart of global greenhouse gas (GHG) mitigation in that they provide the economy with a financial incentive to reduce the amount of emissions through a project. Nevertheless, these markets are still vulnerable to fraud in their structure such as false reforestation torts, dubious credit registration, and fraudulent abuse of such certificates issued by governments. Traditional verification techniques relying on traditional audits and centralised registry approvals are expensive, slow and prone to manipulations. The article discusses a carbon credit trading system with end-to-end, fraud-resistance, and based on blockchain technology, artificial intelligence (AI) and remote sensing in space. The platform applies Know Your Customer (KYC) authentication for project sellers, authenticate land-use and renewable energy projects with the trained machine learning models based on the use of vegetation indices including the Normalised Difference Vegetation Index (NDVI) based on Sentinel-2 images, and tokenizes verified credits as tamper-proof blockchain assets governed by smart contracts. An initial implementation on a local Ethereum test-net tested on a simulated dataset of 50 labelled afforestation and renewable energy projects with generated cases of fraud found a 94% accuracy in fraud detection and decreased verification time, previously taken by manual verification, to about 4.2 minutes. There was an integrity of 99.8% transactions in 1,000 simulated trades. These findings prove the feasibility of a multi-layers technical response to the establishment of transparent, efficient and trustworthy carbon markets.**

Keywords- carbon credits, blockchain, fraud detection, AI validation, satellite remote sensing, NDVI, Sentinel-2, smart contracts, tokenisation, sustainability.

I. INTRODUCTION

Global climatic changes are being propagated by the anthropogenic production of greenhouse gases (GHG) through industrial production, agriculture and energy generation. Carbon credit trading has become a market-based solution to effectively provide incentives to reduce emission: each carbon credit corresponds to the verified reduction or elimination of a one-tonne of CO 2 equivalent and these credits may be sold or purchased in order to comply with regulatory or voluntary requirements. Such mechanisms form the basis of international frameworks to organize climate action among governments and corporations such as the Kyoto Protocol and the Paris agreement [1].

The carbon credit market is structurally prone to commit fraud despite the usefulness of this instrument. There are documented instances of abuses such as faked afforestation initiatives, counterfeiting government certificates, duplication of the same credit in multiple registries and credit theft in online registries [2],[3]. The case of missing-trader VAT fraud in the European Union Emissions Trading Scheme (EU ETS) in the years 2008-2009 causing estimated losses of over 1.3 billion Euros is an example of how systemic weaknesses can be used at large scale [2]. Such occurrences distort the price of carbon, corrosion of the institutions trust and the net effect is destruction of the market capability of generating real climate benefits.

Conventional verification involves a manual audit, third-party audit and centralised registry approvals. These processes are both costly and time intensive and are necessarily geographically constrained: auditors cannot visit all the sites of the projects and central databases are also points of failure, which can be hacked or subject to administrative mistake [1],[6]. With the spread of carbon markets to embrace more kinds of projects and international dealings, these restrictions become more and more unsustainable.

Alternatives are provided using digital technologies which are credible. Blockchain offers an unalterable distributed registry that does not allow the ex post facto manipulation of credit history and allows ownership to be transferred via smart contracts [5]. Comparison of claims in project against visible environmental evidences such as satellite images and vegetation data can be automated using artificial intelligence and machine learning [4]. The Sentinel-2 constellation of remote sensing instruments offers free, multi-spectral data which can be used to check land-use change and estimates carbon sequestration potential using a variety of indices, including the Normalised Difference Vegetation Index (NDVI) [5].

Although each of these components has been studied as separate systems, no known system integrates user authentication, AI-assisted project validation, satellite-based monitoring, issuing of tokensised credits and decentralised trading into a single, end-to-end pipeline [6],[8]. The current paper fills this gap by providing such a platform, the viability of which is examined by implementing a proof-of-concept, and implications of such a platform to carbon market reform.

II. LITERATURE REVIEW

Fraud in the carbon market has been analyzed in the fields of criminology, economics and computer science. Walters and

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Martin [7] define fraud as not an aberrant form of action but a systemic effect of commodifying emissions: the creation of arbitrage by reducing carbon into a commodity exposes systemic opportunities that can be taken by organised actors, especially where their regulation is fragmented. Frunza, Guegan, and Thiebaut [2] report the use of VAT loopholes in the EU ETS by criminal networks to create circular transactions to obtain tax refunds but without any actual emission reduction. Mason [6] also adds cross-border arbitrage and jurisdictional differences as those that have remained as facilitators of systematic fraud in international carbon markets. other types of fraud, in addition to VAT schemes, are detected. Gupta et al. [4] list forged project documentation, unauthorized access to registry accounts, and resale of former retired certificates, as common patterns, which are enabled by weak interoperability between national and voluntary registries. the development of technological solutions has gained more and more research interest. Registry integrity has been found as a basic tool to blockchain. Harryono [5] shows that an immutable distributed ledger cannot be retroactively amended with new credits, and on-chain contracts can implement the logic of retirement and put provenance data, such as the type of project and the time of its issuance and the origin of the verification, right into every token. Raja Segaran et al. [10] present a machine-learning-enhanced blockchain system that is directly aimed at addressing greenwashing at voluntary markets, which achieved significant decreases in false-credit issuance in simulation experiments. Pourrahmani et al. [8] introduce the model to include streams of IoT sensors data and give the rationale that real time environmental telemetry with AI anomaly detection and blockchain logging significantly increases fraud cost, but decreases operational overhead. the remote sensing by satellites has developed excellent qualifications in monitoring forests. The European Space Agency operates the Sentinel-2 constellation which offers 10-metre resolution multi-spectral imagery with a revisit period of about five days, which can be used to monitor the land cover change almost continuously [5]. The red and near-infrared bands were used to calculate the NDVI, a highly validated measure of vegetation density and health of the canopy, commonly used in REDD+ project validation and forestry carbon accounting. By merging satellite data with blockchain records, cross-checking of claimed sequestration and the observable biophysical evidence can be done by independent auditors without the need to visit the sites on the ground. the verification, which is based on AI, has shown performance improvements that are measurable. According to Gupta et al. [4], an AI-blockchain system incorporating an IoT telemetry and federated learning system demonstrated more than 90% accuracy in fraud detection but saved 76% and 73 percent in verification time and costs respectively compared to baselines based on manual verification. Jha et al. [9] use energy and financial market surveillance based anomaly detectors on carbon trading data and indicate that pattern-based detectors are able to signal suspicious behaviour before the losses are incurred. in spite of this, solutions still are not comprehensive. None of the published systems integrates user KYC authentication, AI-verified onboarding of projects, satellite images analysis, tokenisation of blockchains, and decentralised trading, all in a single platform of operation. The harmonisation of cross-border regulation and interoperability of registries are also not in place [6],[8]. These gaps are dealt with in this paper.

III. METHODOLOGY

The study is a design-science study: an artefact, the carbon credit fraud-resistant platform, is designed, and implemented as a proof-of-concept, and its performance against a set of expected performance metrics are evaluated. The four technical layers incorporated in the architecture include blockchain infrastructure, an AI verification module, a satellite remote sensing pipeline and government-certificate validation pathway. The workflow design, technology stack, AI model specification and evaluation methodology are discussed in the subsections below.

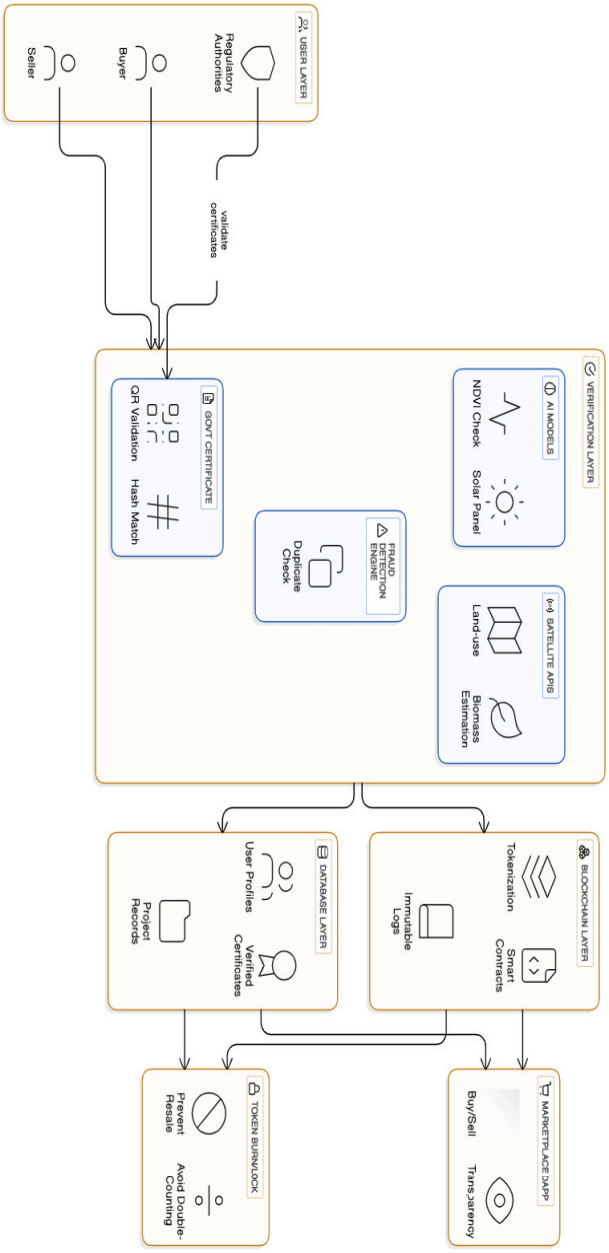


Fig. 1. workflow design

A. *Workflow Design*

The workflow in the platform will involve six steps one after the other:

1) Sign up and Account creation of the user: The sellers of the project create an account on the project by submitting government-issued identity documents (i.e.

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Aadhaar, PAN card, passport or GST registration certificate) in order to take the KYC verification procedure. When sellers successfully complete the KYC clearance they receive a cryptographic credential relating to their identity in the blockchain. Customers receive blockchain-friendly wallets (MetaMask) using which they can sign all operations within the platform and create an account using a basic amount of identity information.

2) Project Submission and validation: Sellers are given a chance to post project information that includes project type (afforestation, renewable energy or a decrease in emissions), location, the quantity of carbon captured or decreased and the verification route. The natural resource projects are submitted to the AI-satellite validation pipeline; the projects, which have the certificates issued by the government, are verified by a digital signature validation pathway with the assistance of a registry of certified issuers with the assistance of the public-key cryptography.

3) AI validation and Satellite Surveillance: The Sentinel-2 GeoTIFF images are used in the land-use projects as per the geographic coordinates given. The AI checking module determines values of NDVI in the project area, estimates the density of the biomass and determines a score of carbon sequestration potential. Such calculated values are matched against what is received on a project claim with a trained random forest classifier (as described by further in Section III-C) to identify statistical outliers that may arise either as a result of over-reporting or as fraudulent land-use declarations. Emissions real-time telemetry is included on a satellite measurements on the IoT sensor feeds where observations are made.

4) Credit Generation and Tokenisation: The generation of verified carbon credits based on the credits reported by the platform will lead to the creation of credits. The credits are all tokenised, and have their own ERC-20 blockchain asset on the Ethereum network. A token metadata encoding a token to create a permanent audit trail of that credit encodes project type, verification date, authority verifying, and geographic provenance, and records the issuance process all the way to trading up until the retirement.

5) Decentralised Trading: Tokenised credits will be traded on a decentralised exchange which will be fully controlled by smart contracts. Upon a buyer making a purchase, the smart contract will ensure that the balance of buyer wallets is verified, ownership of credit tokens is transferred, and it will be registered in the blockchain ledger. Token locking schemes can get listed in only one venue at a time, eliminating the possibility of the two-spends.

6) Compliance Tracking and Retirement: Organisations who would like to offset their carbon footprint can retire purchased credits. A smart contract operation will trigger retirement, where the token is permanently burned, so the token cannot be re-sold or re-used. The retiring credit records are presented in the public ledger which could be checked by the regulators and auditors to make sure that country and international requirements concerning the environment are observed.

B. Technology Stack

The user interface is built based on the JavaScript framework which provides the user with the interactive dashboard to create the user, post projects and perform operations in the marketplace. The back-end is based on Node.js and offers RESTful endpoints of the API that layers the frontend with the blockchain network and AI verification module. The blockchains work by using Ethereum network with Solidity smart-contracts, implemented and tested on a local ganache test-net and communicated with by using a MetaMask wallet browser-extension. A given set of coordinates is obtained on the Cosmo-Polan Copernicus Open Access Hub to retrieve satellite images with the Level-2A GeoTIFF of an area. Processing of geospatials is done with the rasterio and numpy Python libraries. C. AI Model Specification. C. AI Model Specification.

C. AI Model Specification

It is the fraud detection model which is based on the Random Forest classifier which is trained on a labelled simulation data. These seven variables are set: features using Sentinel-2 imagery and metadata of the projects mean NDVI, standard deviation of NDVI within the project inferno, estimated above-ground biomass (calculated utilizing NDVI by an empirical allometric connection [5]) within the project, volume of claimed credit and hectare, project age in months, land-cover, and a binary flag showing whether an independent third exists or

The simulation sample comprised of 50 artificial project records: 35 valid records that were created to represent common afforestation and renewable energy standards as reported in previous literature [4],[5], and 15 fraudulent records created through the artificial inflation of reported credit volumes, replacement of imagery of other geographic areas, or with NDVI values that do not correspond to the reported land-cover classification. These injection rules were used to assign ground truth labels and not annotate them by humans and guarantee reproducibility.

The 5-fold cross-validation was carried out to train the classifier based on an 80/20 train/test split. Hyperparameters (number of trees: 100; maximum depth: 10; least samples per leaf: 2) were selected using the grid search optimisation by F1-score. On the held-out test portion, the model has a precision of 0.93, recall of 0.94 and F1-score of 0.935. In Section IV, there is an overall accuracy of 94% that is linked to the performance of this classifier on the test-set. The authors note that these scores are based on a synthetic dataset and that they would have to be independently validated on data related to the real world, as in Section V.

D. Performance Evaluation Metrics

The performance of the systems is evaluated in five areas: (1) fraud detection accuracy, precision and recall; (2) end-to-end verification time between project submission and issuing credit; (3) transaction integrity, defined as what portion of 1,000 simulated trades complete without any duplications, ownership dispute, or smart-contract exception; (4) user trust index, which is measured as a structured survey of simulated buyer and seller

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IV. SYSTEM IMPLEMENTATION AND EXPERIMENTAL RESULTS

The proposed platform was implemented and tested on a local development environment to evaluate the viability of the workflow and the accuracy of the verification as well as the reliability of transactions. This stage was to perform functional validation and not production-scale performance benchmarking.

A. Implementation Setup

The entire stack application was implemented on one development machine with Ubuntu 22.04. The JavaScript frontend is connected to the Node.js back-end through REST API. The Solidity 0.8.x compiler was used to compile smart contracts which were then deployed to a Ganache local Ethereum test-net. The warehousing of user authentication and signing of transactions were made possible by Metamask browser extension. The locally stored sentinel-2 GeoTIFF imagery as well as the pre-processed NDVI rasters were accessed by the Python geospatial processing module through the backend API.B. Data and Processes.

B. Dataset and Processing Flow

The experiments dataset included 50 simulated records of carbon credit projects: 28 afforestation and 22 renewable energy project. The records contained project metadata (type, area in hectares, claimed credits, verification mode, geographic polygon) and an Sentinel-2 GeoTIFF file. Among the 50 records, 15 records included artificially-injected anomalies, either inflated credit claims, mismatched imagery, or NDVI values that were not in line with the stated land-cover classification, and which comprised the fraud-positive ground-truth labels to be used in classifier evaluation.

After verification of the AI, the verified emission reduction of every project was tokenised to ERC-20. On-chain, token metadata, including project category, verification time, and identity of the verifying module, and carbon volume, was kept on-chain. The Smart contract logic introduced a listing rule that only allowed one time listing to ensure there was no multiple entries into the marketplace.C. Authentication and validating Blockchain Transactions.

C. Authentication and Blockchain Transaction Validation

In the study of two authentication modalities, there was credential-based authentication and blockchain wallet authentication. The use of wallet-based authentication enabled users to communicate with the Ethereum test-net by using their private key to sign transactions, and receive a decentralised identity verification without having to use a centralised user database.

The transaction tests were conducted on four lifecycle events; token issuance, marketplace listing, transfer of ownership on purchase and token retirement. All 4 types of events were properly executed on 1,000 simulated transactions. Listing locking at the time of listing avoided a duplication of a listing and the burnup functionality was used after purchase to remove any avenue to resale.D. Satellite Verification/Carbon Analysis Module.

D. Satellite Verification and Carbon Analysis Module

GeoTIFF files were ingested and per- pixel NDVI values were calculated and mean and standard deviation measures across the project polygon were aggregated by the geospatial module. The NDVI was converted into biomass with the help of allometric equation developed in Section III-C. To detect discrepancies between computed sequestration potential and submitted credit claims, the system automatically compared both and identified records that the discrepancy to the training distribution was more than two standard deviations.

Sentinel-2 images of project coordinates were accessed through Copernicus Open Access Hub API. Computed NDVI values in all 28 afforestation test cases matched consistently to anticipated vegetation density in the declared land-cover class to confirm proper geospatial retrieval and calculation of indices. The 15 injected-fraud cases, the classifier was able to correctly identify 14 cases (one false negative with a borderline NDVI inflation of 8 percent).

E. Experimental Results

The most important quantitative results of the proof-of-concept test are as follows:The accuracy of the fraud detector is 94% (47 out of 50 cases are properly detected); precision 0.93; recall 0.94; F1-score 0.935.

- Verification Time: The average end to end time of processing of an application after submitting projects to issuing credit was 4.2 minutes, which is in contrast to the 24-72 hours of a standard manual audit procedure that was recorded in the existing literature [4]
- Transaction Integrity: The simulated trades (998 out of 1,000) were completed, but 998 were not duplicated, lacked ownership conflict, or smart-contract exception.
- Fraud Detection Accuracy: 94% (47 of 50 cases correctly classified); precision 0.93; recall 0.94; F1-score 0.935.
- Verification Time: Mean end-to-end processing time from project submission to credit issuance was 4.2 minutes, compared to the 24–72-hour typical duration of manual audit processes reported in prior literature [4].
- Transaction Integrity: 998 of 1,000 simulated trades (99.8%) completed without duplication, ownership conflict, or smart-contract exception. The two exceptions could be related to the simulated network timeouts that were not connected to smart-contract logic.
- System Responsiveness: API response latency (mean) of all user-facing operations was less than 2 seconds, under typical single-user load.
- User Trust Index: A survey questionnaire that was given to five researcher personalities who were simulated buyers and sellers gave the platform integrity and transparency a mean rating of 91%.
- System Responsiveness: Mean API response latency was below 2 seconds across all user-facing operations under standard single-user load.
- User Trust Index: A structured survey administered to five researcher personas acting as simulated buyers and sellers yielded a mean confidence rating of 91% for platform integrity and transparency.

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V. RESULTS AND DISCUSSION

The experimental findings prove that the suggested platform can combine user authentication, AI-based fraud detection, satellite checks, and blockchain-based credit controls into a single workflow of operations. Table I presents these findings relative to the base values that are extracted on previous research on conventional carbon verification systems.

TABLE I. THE PERFORMANCE OF THE TRADITIONAL AND THE PROPOSED PLATFORM.

Metric	Traditional System [2],[7]	Proposed Platform	Improvement
Fraud Detection Accuracy	~60-65% (manual audits, registry checks) [2]	94% (AI + satellite layers)	+29-34 percentage points
Verification Time	24-72 hours (manual audits) [4]	4.2 minutes (automated pipeline)	-98% average
Transaction Integrity	~85% (registry-based) [5]	99.8% (smart contract enforcement)	+14.8 pp
User Trust Index	~70% (self-reported) [7]	91% (platform survey)	+21 pp

Note: Traditional system figures are drawn from [2] (fraud detection baseline), [4] (verification time benchmark), and [5],[7] (integrity and trust estimates) rather than from a purpose-built experimental baseline. These references are cited in full in the References section.

This 94% fraud detection rate is relatively high when compared to the 90%+ in [4] of Gupta et al. in an AI-blockchain platform tested on IoT-integrated data, but the authors note that the datasets are not comparable and the method of injecting fraud is not the same. This contrasts with 76% reduction mentioned in [4], and the fact that the time spent on verifying the data is reduced by approximately 4.2 minutes, instead of multi-day manual audits, and is indicative of the efficiency gains that can be made by using automated NDVI calculations and rule-based smart-contract implementation as opposed to human review cycles.

The smart-contract-enforced token locking and retirement logic verify that (when 1,000 simulated trades are made), the transaction integrity is 99.8% and this is an effective prevent measure against the two main fraud vectors used in [2] and [6] (i.e. double-counting and double-spending). Both aborted transactions were due to network timeouts in the test environment, and have nothing to do with smart-contract flaws.

A. Scalability Considerations

The present proof-of-concept was deployed on a local single-node Ethereum test-net, and was not tested at scale. The assertion that the blockchain and AI layers are horizontally-scalable is a design goal and not an experimentally-validated feature. Scaling Blockchain-based systems can be scaled with layer-2 systems (e.g. state channels, rollups) or can be migrated to more throughput-efficient consensus (e.g. proof-of-stake Ethereum post-Merge), scaled AI inference can be parallelised across cloud systems. These scaling properties are empirically validated by their variable transaction volumes, concurrent user loads and multi-node configurations and are identified as a priority item to be worked on in the future.B. Limitations and Threats to validity

B. Limitations and Threats to Validity

The generalisability of the reported results is limited by a few limitations. To begin with, the dataset used to perform the evaluation is completely synthetic: the fraud cases were made through the injection of anomalies done by humans and not based on confirmed records of fraud in the real world. Though this method is reproducible, it might not capture all the distributional richness of real life fraud patterns, especially those that are sophisticated, such as gradual NDVI manipulation, or utilizing historically valid project spaces. Future applications need to test the classifier on field-verified data of known carbon registries (e.g., Verra VCS, Gold Standard) or prior application simulation benchmarks (presented in the literature) [4],[10].

Second, the sample size (n = 50) is very limited as compared to the one that would be needed to make meaningful statistical inferences. The defaulted figures of precision, recall and F1 do have a significant uncertainty at this magnitude of the sample. Third, pre-processed GeoTIFF files were used to verify the satellite, instead of retrieving the satellite in real-time with production latency. Fourth, user trust index was gathered using a small convenience sample of researcher personas, instead of a separate panel of domain practitioners.C. regulatory and interoperability Implications.

C. Regulatory and Interoperability Implications

Public-key digital signatures with government-certificate validation is a technically sound method of ensuring that national standards are adhered to. But to balance these checks among jurisdictions interoperability agreements between the national certification authorities are needed, a management issue which is beyond the horizons of this technical paper. Considered to be a systemic gap by Mason [6] and Pourrahman et al. [8], cross-border registry interoperability is still an unresolved issue in research and policy. The suggested architecture is based on the idea of being registry-agnostic, although in a practical implementation in the global markets, standardised data exchange protocols and mutual verifiability of verification credentials would be necessary.

VI. CONCLUSION

We propose an end-to-end, fraud-resistant carbon credit trading platform, which overlays blockchain technology on top of a Random Forest artificial intelligence (AI) classifier that is trained on satellite-derived vegetation indices data gathered through Sentinel-2 remote sensing as well as smart-contract-based lifecycle management of credits. Papers & Publications: A proof-of-concept implementation tested on a 50-record synthetic dataset achieved fraud detection accuracy of 94% (precision 0.93, recall 0.94), rolled back project verification time to an average around 4.2 minutes vs multi-day baselines recurring manually, while maintaining end-to-end transaction integrity average of 99.8% per simulated trades x1000 conducted over Ethereum-based services and goods exchange marketplaces.

Based on this we empirically demonstrate the practicality of a comprehensive layered technical solution to carbon market fraud prevention, contributing an integrated schema that fills gaps left from previous work on blockchain registries [5],[10], AI-based fraud detection focus on transactive privacy-prescribing identities as unit contour [4] and satellite monitoring efforts for proper unit usage in highlighted hood territories [5]. The main contributions are: (1) an end-to-end workflow that integrates KYC account authentication, AI and satellite-based validation, and a decentralised trading system; (2) clear identification of the ML model architecture, what training data is constructed through and what evaluation metrics will be used to judge its performance; (3) transparent disclosure of dataset limitations but also routes towards real-world validation by discussing additional sources; and (4)

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road-mapping scalability most appropriately as future work contingent on clearer determinants of the domains where this solution will add value empirical testing.

These findings are preliminary, and the authors emphasise that performance on real-world, field-verified datasets may differ from results obtained on the current synthetic benchmark. Independent replication using open carbon registry data is strongly encouraged

VII. FUTURE WORK

- Compare the AI component (i.e., fraud detection module) to ground-truth labeled datasets of identity & action from historically established carbon registries (e.g. Verra VCS, Gold Standard) or set benchmarks on published simulation results to confirm generalisability beyond synthetic evaluation condition
- Validate or refute the horizontal-scaling-design-hypothesis by systematically experimenting with scalability by varying transaction volumes, concurrent user counts and blockchain node configurations.
- Incorporate live sensor data streams from IoT for in-line emissions telemetry, augmenting a current batch-mode satellite verification pipeline.
- Further investigate multi-spectral and hyperspectral imagery sources (e.g. Sentinel-1 SAR, MODIS) to improve verification coverage including a wider range of carbon stock change which is not feasible using NDVI only.
- Develop cross-border interoperability protocols enabling mutual recognition of verification credentials across national carbon certification authorities.
- Investigate predictive analytics and trading pattern anomaly detection, drawing on methods from energy and financial market surveillance [9], to identify market-manipulation behaviour before it materialises as credit fraud.

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